Due Date

You must *submit* the source code for the solution to this lab exercise to *Moodle* by

Monday, October 20, 2025

in order to receive full credit for this work. You must also *demonstrate* the solution to the instructor during class, at the earliest opportunity.

Lab Exercise

The Stock class: Following the examples of the

CircleWithPrivateDataFields and
TestCircleWithPrivateDataFields classes

(provided on *Moodle*), design a class named Stock that contains:

- A String data field named **symbol**, for the stock's symbol.
- A String data field named **name**, for the stock's name.
- A double data field named **previousClosingPrice** that stores the stock price for the previous day.
- A double data field named **currentPrice** that stores the stock price for the current time.
- "Getter" and "Setter" methods for each of the data fields (as needed):
  getSymbol(), **setSymbol()**,
  getName(), **setName()**,
  getPreviousClosingPrice(), **setPreviousClosingPrice()**,
  getCurrentPrice(), **setCurrentPrice()**.

  Note: Depending on the details of your design, you may or may not need all of these "getter" and "setter" methods. Implement those that you find necessary to make your solution work correctly.
- A constructor that creates a stock with a specified symbol and name.
- A method named **getChangePercent()** that returns the percentage changed from **previousClosingPrice** to **currentPrice**.

Write a test program that prompts the user for a stock symbol. If the user enters "q", then the program must exit. If the user enters anything other than "q", then the program must assume that the input represents a stock symbol.

After a stock symbol is input, the program must prompt the user for the company name associated with the stock symbol, and also the company's previous closing price and current price. The program must create an instance of the Stock class, and utilize the appropriate set methods to set the object's internal field values. After the object is created and its member variables are set, the program must call the **getChangePercent()** method to determine how much the stock price has changed.

When this processing is finished for one stock, the main loop must ask the user for another stock symbol, and repeat the process. The program exits when the user enters the string "q" in place of a stock symbol.

Your program must have two classes: a public class named **Lab9a**, and another (non-public) class named **Stock**. The **Lab9a** class contains a **`main()`** method, which creates an object of the **Stock** class.

HINT:

You will need to create a **`Scanner`** object to input the various data values from the console. You may find difficulties if your program uses both **`Scanner.nextLine()`** and **`Scanner.nextDouble()`** methods:

- On repeated passes through the main loop, the program appears to skip over the statement that inputs the stock symbol.
- One way to avoid such problems is to use **`Scanner.nextLine()`** for all inputs, and then convert the input strings to double values by calling **`Double.parseDouble()`** where necessary.

Sample Output

In the sample output shown, text that the user types is shown in **BOLD** font. When the program actually runs, all text is shown in the same font.

```
Enter stock Symbol (or 'q' to exit): ORCL
Enter company name: Oracle Corporation
Enter previous closing price: 88.90
Enter current price: 97.50
SYMBOL=ORCL, NAME=Oracle Corporation, PreviousPrice=88.90, CurrentPrice=97.50,
PercentChange= +9.67 %

Enter stock Symbol (or 'q' to exit): IBM
Enter company name: International Business Machines
Enter previous closing price: 134.22
Enter current price: 111
SYMBOL=IBM, NAME=International Business Machines, PreviousPrice=134.22,
CurrentPrice=111.00, PercentChange=-17.30 %

Enter stock Symbol (or 'q' to exit): q
Exiting program...
```

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